

Feedforward 4D Gaussians for Egocentric Vision

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Abstract

While 3D Gaussian Splatting (3DGS) has become the standard for real-time radiance field rendering, extending these models to dynamic, egocentric environments in a feedforward manner remains challenging. Current 4DGS methods rely on per-scene optimization and do not exploit egocentric cues such as hand-object interactions. We extend NeoVerse, a feedforward 4DGS model, with a HaMeR-style head that estimates a parametric hand model (MANO), and inject its features into the Gaussian head to guide and supervise the reconstruction. The model jointly estimates camera poses, hand poses, and pixel-aligned Gaussians. On the HOT3D dataset, hand-feature injection improves full-frame Gaussian PSNR from 36.96 to 37.92 dB over a finetuned NeoVerse baseline. Our hand head reaches 5.71 mm PA-MPJPE, down from 5.93 mm without local-global fusion. These scene metrics are computed over the full frame and understate the hand-region gain. We release our training and evaluation code.

1. Introduction

3D Gaussian Splatting (3DGS) [8] has become a standard explicit scene representation for high-quality 3D reconstruction. It models scenes as 3D Gaussians optimized by differentiable rendering. It reaches the quality of implicit approaches such as Neural Radiance Fields (NeRF) [9] while training efficiently and rendering in real time. Feedforward 3DGS methods [2] avoid per-scene optimization and estimate the Gaussians directly from input images. 4D Gaussian Splatting (4DGS) [14] extends 3DGS to dynamic scenes with a temporal dimension. Recent geometry foundation models such as DUST3R [13] and VGGT [12] predict pose, depth and dense geometry from raw video in one pass. NeoVerse [15] builds on a VGGT backbone and emits a 4D Gaussian field from monocular video.

At the same time, egocentric video has gained increasing

attention through large-scale datasets such as Ego4D [6] and capturing devices like Project Aria [4]. Egocentric data poses specific challenges. The hands are prominent in the frame, articulate quickly, self-occlude, and interact with objects. A hand-agnostic 4D world model does not represent this structure and reconstructs the hand region worst.

The combination of feedforward 4DGS and egocentric data is largely unexplored, and current pipelines do not use cues such as hand pose. We ask: does an explicit hand prior injected into a feedforward 4D world model improve egocentric scene reconstruction, and by how much? We integrate a HaMeR-style [10] head that predicts MANO [11] parameters into NeoVerse, inject its features into the Gaussian head, and measure the change on HOT3D [1].

Our contributions are:

- We integrate a HaMeR-style MANO hand head into the NeoVerse 4D Gaussian model and inject its features into the Gaussian head.
- We show that this hand prior improves full-frame scene reconstruction on HOT3D. PSNR rises by 0.96 dB and LPIPS drops from 0.005 to 0.004 over a finetuned baseline.
- We find that local-global fusion between hand crops and scene features drives hand accuracy. It lowers PA-MPJPE from 5.93 to 5.71 mm, and a variant that detaches the hand from the scene gradient is worse on all four metrics.
- We report full-frame scene metrics and explain why region-masked evaluation is needed to measure the hand-region effect (Sec. 4.3).

2. Related Work

Feedforward and dynamic Gaussian Splatting. 3DGS [8] and its dynamic extension 4DGS [14] achieve high fidelity but require per-scene optimization. Feedforward methods such as pixelSplat [2] amortize this by predicting pixel-aligned Gaussians directly, trading per-scene tuning for a learned prior. We build on this

feedforward setting through NeoVerse and study what a hand-specific prior adds.

Geometry foundation models and 4D world models. DUST3R [13] reframed two-view geometry as direct pointmap regression, and VGGT [12] scaled this to a transformer that predicts cameras, depth and tracks jointly. NeoVerse [15] uses a VGGT backbone to produce a 4D Gaussian world model from monocular video. These models are general and have no notion of hands. We keep the backbone frozen and add a hand head and a feature-level prior instead of retraining it.

Monocular hand reconstruction. The MANO model [11] provides a differentiable, low-dimensional parameterization of hand pose and shape that is the de facto target for hand estimation. HaMeR [10] showed that a plain transformer, trained at scale, regresses MANO parameters from a single image robustly. We reuse the HaMeR formulation but recast it as a head on shared backbone features instead of a standalone network. The hand estimate and the scene then share representation and supervision. Egocentric hand-object datasets such as ARCTIC [5] and HOT3D [1] supply the 3D hand and camera annotations needed to train such a head.

Concurrent hand-scene reconstruction. Closest to our setting is the concurrent Hand3R [7], which reconstructs hand meshes and scene geometry online from monocular video by pairing a hand expert with a 4D scene foundation model. We share the high-level goal of jointly recovering hands and scene, but differ in construction and objective. Hand3R targets online 4D hand-scene geometry. We instead target a feedforward *Gaussian* radiance field and use the hand estimate as a feature-level prior injected into the Gaussian head. We supervise it jointly with photometric reconstruction, rather than fusing two reconstructed geometries. The two are independent and complementary.

3. Method

3.1. Overview and backbone

The input is a monocular egocentric clip $\mathbf{I} \in \mathbb{R}^{T \times 3 \times H \times W}$ of $T=16$ frames at $H=W=224$, with per-frame intrinsics $K_t = (f_t, c_{x,t}, c_{y,t})$. Per frame t the model predicts (i) a pixel-aligned 4D Gaussian field $\{(\mu_k, \Sigma_k, o_k, c_k)\}_k$ and a depth map $D_t \in \mathbb{R}^{H \times W}$, (ii) the camera pose, and (iii) for each visible hand $h \in \{L, R\}$ the MANO [11] parameters $\mathbf{m}_{t,h} = (\mathbf{t}_{t,h}, \mathbf{q}_{t,h}, \boldsymbol{\theta}_{t,h}, \beta_{t,h}) \in \mathbb{R}^{32}$: a translation $\mathbf{t}_{t,h} \in \mathbb{R}^3$, a global rotation $\mathbf{q}_{t,h} \in \mathbb{R}^4$ (quaternion), a reduced pose $\boldsymbol{\theta}_{t,h} \in \mathbb{R}^{15}$, and a shape $\beta_{t,h} \in \mathbb{R}^{10}$. We build on NeoVerse [15], whose VGGT backbone [12] produces

multi-level dense features that are decoded by DPT-style heads into camera, depth and Gaussian parameters. Each pixel emits a Gaussian (mean, covariance, opacity and color), and a frame is rendered by differentiable alpha compositing as in 3DGS [8], using the gsplat rasterizer. We keep the backbone *frozen* (no gradient flows into it) throughout, both to preserve the geometry prior learned at scale and to keep training tractable on a single GPU. All new capacity lives in the hand head and in the injection pathway (Fig. 1).

3.2. Multi-hand MANO head

MANO [11] maps pose $\boldsymbol{\theta}$ and shape β to a hand mesh by linear blend skinning,

$$\mathbf{V}(\boldsymbol{\theta}, \beta) = W(\bar{\mathbf{T}} + B_S(\beta) + B_P(\boldsymbol{\theta}), J(\beta), \boldsymbol{\theta}, \mathcal{W}), \quad (1)$$

with $V=778$ vertices $\mathbf{V} \in \mathbb{R}^{778 \times 3}$, where $\bar{\mathbf{T}}$ is the template mesh, B_S and B_P are the shape and pose blend shapes, $J(\beta)$ regresses joint centers, and W is the skinning function with weights $\mathcal{W}$. The $J=21$ joints are $\mathbf{J} = \mathcal{R}\mathbf{V} \in \mathbb{R}^{21 \times 3}$ for a fixed regressor $\mathcal{R}$. Following HaMeR [10], our head is a transformer decoder that attends to backbone features and emits, for each hand h , the parameters $\mathbf{m}_h = (\mathbf{t}_h, \mathbf{q}_h, \boldsymbol{\theta}_h, \beta_h)$, with the metric translation $\mathbf{t}_h$ placing the joints in camera coordinates as $\mathbf{X}_h = \mathbf{J}(\boldsymbol{\theta}_h, \beta_h) + \mathbf{t}_h$. We supervise them in 2D by projecting into the image with the known camera intrinsics K . This differs from HaMeR, which regresses a weak-perspective crop camera. Because our hands are placed metrically in the camera frame, we reproject with the real intrinsics. A single decoder pass produces both hands, conditioned on *local-global fused* features: the per-hand crop tokens cross-attend to the global scene tokens of the backbone, so each hand prediction sees the surrounding scene and the other hand. The fusion ablation (Sec. 4.2) isolates this as the dominant design choice.

3.3. Feature-space cropping

A hand occupies a small fraction of an egocentric frame, so processing the full image through the head wastes computation. Rather than re-encoding a pixel crop, we crop in *feature* space: given a per-frame hand bounding box b_h (rescaled by a fixed factor to add context), we pool the backbone feature map inside b_h to obtain the hand tokens that condition the decoder. This reuses the single backbone pass already computed for the scene and avoids a second forward over a pixel crop, which is the dominant cost in a naive HaMeR-on-crops pipeline.

3.4. Hand-to-Gaussian injection

The hand head produces, per hand, enhanced crop tokens $\mathbf{E}_h \in \mathbb{R}^{K \times C}$ ($C=1024$) that have cross-attended to the

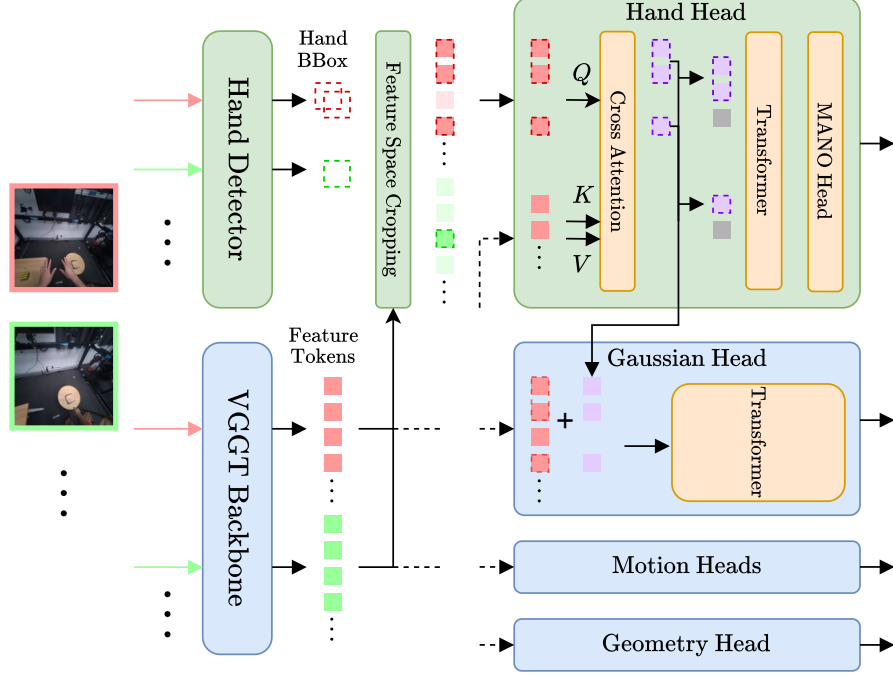


Figure 1. **Architecture overview.** A frozen VGGT backbone, inherited from NeoVerse [15], encodes the egocentric clip. We add a HaMeR-style [10] hand head that regresses MANO parameters for both hands from feature-space crops, and we inject its intermediate hand-aware tokens into the Gaussian head’s feature pyramid (green modules are added by us). The camera, Gaussian and hand heads are trained jointly while the backbone stays frozen.

global scene context and now carry an explicit hand prior. For each level i of the Gaussian head’s DPT pyramid $\mathbf{F}_i \in \mathbb{R}^{C_i \times h_i \times w_i}$ ($C_i \in \{256, 512, 1024, 1024\}$), a 1×1 convolution W_i maps the tokens to C_i channels and we add $\text{resize}_{b_h}(W_i \mathbf{E}_h)$ into the box region b_h of $\mathbf{F}_i$, for each valid hand. The Gaussian head then reconstructs the hand region conditioned on hand location and pose, instead of inferring it from appearance. This injection is the only path from the hand prior to the radiance field. The GS ablation toggles it.

3.5. Training objective

The backbone is frozen. We train the camera, Gaussian and hand heads jointly with a sum of photometric and hand terms,

$$\mathcal{L} = \underbrace{\lambda_1 \mathcal{L}_{L1} + \lambda_2 \mathcal{L}_{LPIPS}}_{\text{Gaussian}} + \underbrace{\lambda_3 \mathcal{L}_{\text{MANO}} + \lambda_4 \mathcal{L}_{3D} + \lambda_5 \mathcal{L}_{2D}}_{\text{hand}}. \quad (2)$$

The Gaussian terms are an ℓ_1 photometric loss and a perceptual LPIPS [16] loss between rendered and ground-truth frames. $\mathcal{L}_{\text{MANO}}$ is an ℓ_1 loss on the predicted parameters $\mathbf{m}_{t,h}$ (translation, rotation, pose and shape, masked per hand). $\mathcal{L}_{3D}$ is a pelvis-aligned error on the joints $\mathbf{J}_{t,h}$. $\mathcal{L}_{2D}$ is the 2D reprojection error of the predicted joints against the annotated keypoints. We train with clips

of 16 frames at 224×224 , AdamW, bf16 autocast, and gradient accumulation.

4. Experiments

Dataset. We train and evaluate on HOT3D [1], an egocentric hand-object dataset captured with Project Aria. Each sequence is a first-person recording of a person manipulating objects, with high-fidelity 3D MANO hand annotations and calibrated camera tracks. We use 198 sequences, sliced into non-overlapping 16-frame clips, and hold out sequences for validation and test. All metrics are computed on held-out sequences.

Metrics. For Gaussian reconstruction we report PSNR, SSIM and LPIPS [16] between rendered and ground-truth frames. For hand estimation we report Procrustes-aligned vertex and joint error (PA-MPVPE, PA-MPJPE, in mm) and the corresponding area-under-the-curve of correct keypoints (AUC-V, AUC-J), the standard HaMeR protocol.

Baselines. For scene reconstruction we compare against NeoVerse in two settings: the *frozen* pretrained model run on HOT3D, and a *trained* variant whose heads we finetune on HOT3D without any hand prior. Our model adds hand

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeoVerse (frozen)	35.75	0.970	0.008
NeoVerse (trained)	36.96	0.977	0.005
Hand Inj. (Ours)	37.92	0.977	0.004

Table 1. **Gaussian reconstruction on HOT3D.** Hand-feature injection on top of the finetuned NeoVerse baseline improves PSNR and LPIPS, with SSIM on par. Higher is better for PSNR/SSIM, lower for LPIPS.

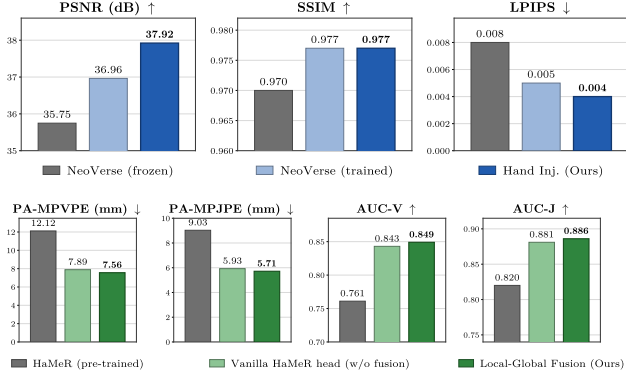


Figure 2. **Ablations.** Top: Gaussian reconstruction. Hand injection improves PSNR and LPIPS over the frozen and finetuned NeoVerse baselines. Bottom: hand head. Local-global fusion (cross-attention between hand crops and global scene features) gives the best hand-mesh accuracy and far surpasses off-the-shelf HaMeR.

injection to the trained variant, which separates the effect of the prior from that of finetuning. For hands the controlled comparison is internal: our retrained head *without* local-global fusion versus the full head *with* fusion, plus a decoupled variant. All are trained on HOT3D under one protocol, so only the design choice varies. We additionally list off-the-shelf HaMeR [10] *for reference only*. It was not trained on the HOT3D egocentric viewpoint, so the gap to it mixes domain adaptation with architecture and is not a clean method comparison.

4.1. Gaussian reconstruction

Table 1 reports the scene results. Finetuning the NeoVerse heads on HOT3D lifts PSNR from 35.75 to 36.96 dB. Hand injection raises it to 37.92 dB (+0.96 dB over the finetuned baseline) and lowers LPIPS from 0.005 to 0.004. SSIM is saturated at 0.977 and does not move. These are *full-frame* metrics. Hands cover a small fraction of an egocentric frame, so the static background dominates the average and understates the gain in the region the prior acts on. Region-masked PSNR and LPIPS are required to measure it (Sec. 4.3). Figure 2(top) plots the gains. Figure 4 shows a reconstructed Gaussian frame.

Method	PA-MPVPE $\downarrow$	PA-MPJPE $\downarrow$	AUC-V $\uparrow$	AUC-J $\uparrow$
HaMeR (pre-trained, <i>reference</i>)	12.12	9.03	0.761	0.820
Ours, head w/o fusion	7.89	5.93	0.843	0.881
Ours, local-global fusion	7.56	5.71	0.849	0.886

Table 2. **Hand reconstruction on HOT3D** (PA-MPVPE / PA-MPJPE in mm). Our controlled result is the fusion ablation (bottom two rows, identical protocol): local-global fusion improves every metric. Off-the-shelf HaMeR is listed for reference only and is not trained on the HOT3D viewpoint, so its gap conflates domain adaptation with architecture.

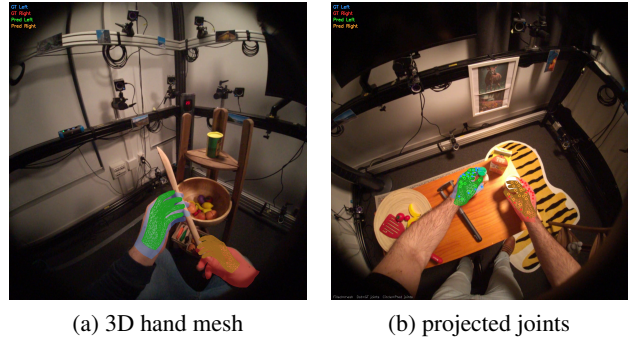


Figure 3. **Hand predictions vs. ground truth.** (a) the recovered 3D MANO mesh. (b) the same prediction with 3D joints projected into the image.

4.2. Hand reconstruction

Table 2 reports the hand results. The main comparison is the fusion ablation, with the same training protocol in both rows. Local-global fusion lowers PA-MPJPE from 5.93 to 5.71 mm and improves the other three metrics. A variant that detaches the hand tokens from the scene gradient is worse on all four metrics, so the gain comes from coupling the hand to the scene. We list off-the-shelf HaMeR (9.03 mm PA-MPJPE) as a reference. It was not trained on the HOT3D viewpoint, so this gap also reflects in-domain training, and we do not read it as a method result. Figure 3 shows example fits and projected joints.

4.3. Discussion

We make two observations. First, the hand prior improves the scene reconstruction itself: full-frame PSNR and LPIPS rise while SSIM stays flat. Second, the hand accuracy comes from fusing local hand crops with global scene context, since the decoupled variant is worse. Both support the hypothesis that a parametric hand estimate is a cheap and useful prior for egocentric 4D reconstruction.

Next measurement. The scene gains above are full-frame and therefore diluted, since the background dominates the average while the prior acts on the hand and object regions.

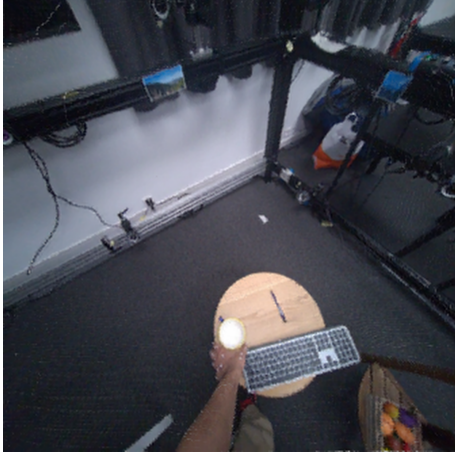


Figure 4. **Gaussian head, qualitative.** A reconstructed Gaussian frame on HOT3D.

The next experiment is to recompute PSNR and LPIPS inside masks of those regions. The pipeline already caches the projected MANO joints and hand boxes, and HOT3D provides object poses and meshes for the object masks. We expect a larger gain in the masked regions. An improvement in the object region would be the stronger result, since it shows that the hand prior helps reconstruct the manipulated object. We report this as the main open direction, not a completed result.

Other limitations. We keep the backbone frozen, which bounds domain adaptation. Parameter-efficient finetuning is a natural extension. Our hand comparison is controlled only internally (the fusion ablation), and a fair external baseline would finetune HaMeR or a similar model in-domain under our protocol. Finally, all results are on HOT3D. Transfer to a second dataset such as ARCTIC [5] or EPIC-KITCHENS [3] would test whether the effect is dataset specific. We also report single-run numbers without variance, which per-sequence statistics should accompany.

5. Contributions, Work Distribution and Code Provenance

Novel contributions. We report two main results. First, a hand prior injected into a feedforward 4D Gaussian head improves the scene reconstruction, not only the hand mesh. Second, the hand accuracy comes from local-global fusion, as the decoupled variant shows. The mechanism is a multi-hand single-pass MANO head on a frozen VGGT/NeoVerse backbone, feature-space cropping, and hand-to-Gaussian injection into the DPT pyramid. To our knowledge this combination has not been used before. The concurrent Hand3R [7] also reconstructs hands and scene, but fuses two reconstructed geometries online, while we inject a hand

prior into a feedforward Gaussian head and supervise it with photometric reconstruction.

Work distribution. The project was carried out by three team members. **Dario Monopoli** led the data pipeline (HOT3D loading, MANO ground-truth caching, hand bounding-box extraction), the training and evaluation infrastructure, and the performance and profiling work that made full-dataset training feasible. **Batuhan Iliev** led the hand head (the multi-hand MANO transformer decoder and the local-global fusion) and the hand-metric evaluation. **Marian Schneider** led the hand-to-Gaussian injection and the Gaussian reconstruction evaluation. Method design, the ablation study, the report and the poster were shared across all three. (Team members should adjust this paragraph to match their own records.)

Code we used vs. code we wrote. We *reused* external code and models: the NeoVerse 4D world model and VGGT backbone [12, 15] with its frozen pretrained weights, the MANO hand model [11], the HaMeR architecture as the starting point for our head [10], the differentiable Gaussian rasterizer of the backbone, and standard LPIPS and rendering-metric implementations. We *implemented ourselves* the multi-hand MANO head and its local-global fusion, the feature-space cropping, the hand-to-Gaussian injection into the DPT pyramid, the HOT3D data pipeline and caching, the training loop and loss (Eq. (2)), and the evaluation and ablation harness for Tab. 1 and Tab. 2.

6. Conclusion

We presented a feedforward 4D Gaussian model for egocentric video with a hand head. We attach a multi-hand MANO head to a frozen VGGT/NeoVerse backbone and inject its features into the Gaussian head. On HOT3D the hand prior raises full-frame scene PSNR from 36.96 to 37.92 dB over the finetuned baseline. Local-global fusion lowers hand error from 5.93 to 5.71 mm PA-MPJPE, and the decoupled variant is worse. Feature-space cropping keeps the added cost low. Region-masked evaluation of the hand and object regions is the next step toward a conclusive result.

References

- [1] Prithviraj Banerjee, Sindi Shkodrani, Pierre Moulon, Shreyas Hampali, Shangchen Han, Fan Zhang, Linguang Zhang, Jade Fountain, Edward Miller, Selen Basol, Richard Newcombe, Robert Wang, Jakob Julian Engel, and Tomas Hodan. HOT3D: Hand and object tracking in 3D from egocentric multi-view videos. *arXiv preprint arXiv:2411.19167*, 2024. 1, 2, 3
- [2] David Charatan, Sizhe Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelSplat: 3D Gaussian splats from image pairs

for scalable generalizable 3D reconstruction. In *CVPR*, 2024. 1

- [3] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, and Michael Wray. Scaling egocentric vision: The EPIC-KITCHENS dataset. In *ECCV*, 2018. 5
- [4] Jakob Engel et al. Project Aria: A new tool for egocentric multi-modal AI research. *arXiv preprint arXiv:2308.13561*, 2023. 1
- [5] Zicong Fan, Omid Taheri, Dimitrios Tzionas, Muhammed Kocabas, Manuel Kaufmann, Michael J. Black, and Otmar Hilliges. ARCTIC: A dataset for dexterous bimanual hand-object manipulation. In *CVPR*, 2023. 2, 5
- [6] Kristen Grauman et al. Ego4D: Around the world in 3,000 hours of egocentric video. In *CVPR*, 2022. 1
- [7] Wendi Hu, Haonan Zhou, Wenhao Hu, and Gaoang Wang. Hand3R: Online 4D hand-scene reconstruction in the wild. *arXiv preprint arXiv:2602.03200*, 2026. 2, 5
- [8] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian splatting for real-time radiance field rendering. In *ACM Transactions on Graphics (SIGGRAPH)*, 2023. 1, 2
- [9] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020. 1
- [10] Georgios Pavlakos, Dandan Shan, Ilija Radosavovic, Angjoo Kanazawa, David Fouhey, and Jitendra Malik. Reconstructing hands in 3D with transformers. In *CVPR*, 2024. 1, 2, 3, 4, 5
- [11] Javier Romero, Dimitrios Tzionas, and Michael J. Black. Embodied hands: Modeling and capturing hands and bodies together. *ACM Transactions on Graphics (SIGGRAPH Asia)*, 2017. 1, 2, 5
- [12] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer. *arXiv preprint arXiv:2503.11651*, 2025. 1, 2, 5
- [13] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jérôme Revaud. DUST3R: Geometric 3D vision made easy. In *CVPR*, 2024. 1, 2
- [14] Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 4D Gaussian splatting for real-time dynamic scene rendering. In *CVPR*, 2024. 1
- [15] Yuxue Yang, Lue Fan, Ziqi Shi, Junran Peng, Feng Wang, and Zhaoxiang Zhang. NeoVerse: Enhancing 4D world model with in-the-wild monocular videos. *arXiv preprint arXiv:2601.00393*, 2026. 1, 2, 3, 5
- [16] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, 2018. 3